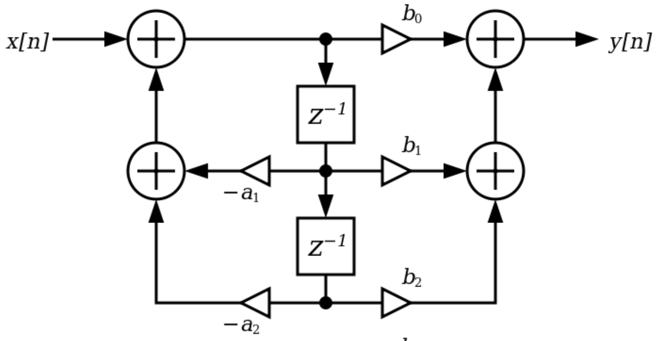
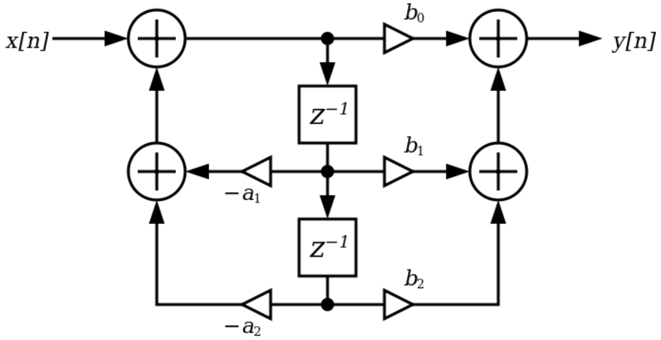




Embedded Signal Processing Systems

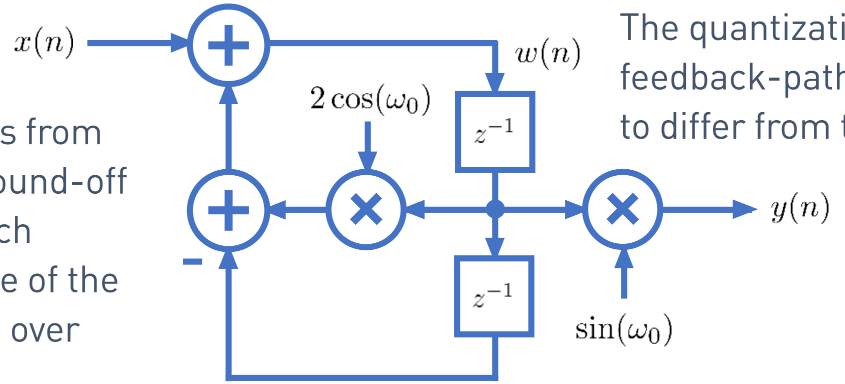
Master of Science

Prof. Dr. C. Jakob
University of Applied Sciences Darmstadt
h_da
Faculty of Electrical Engineering and Information Technology
fbeit

Discrete-time sinusoidal oscillators

- The implementation of this oscillator with finite word-lengths leads to several quantization effects what finally impacts the waveform accuracy …

Round-off noise
Rounding of the output products from the cosine multiplier leads to round-off noise in the recursive loop, which causes the amplitude and phase of the output signal to slightly change over time.



The quantization of the cosine term $\cos(\omega_0)$ in the feedback-path causes the actual output frequency to differ from the ideal frequency ω_0.

Coefficient quantization noise
The quantization of the sine term $\sin(\omega_0)$ in the output-path causes the actual output amplitude to differ slightly from A.

- Round-off noise is potentially the most severe noise-source. It is cumulative, and it therefore requires periodic resetting of the initial conditions in the feedback-path to reinitialize proper amplitude and phase values.

Discrete-time sinusoidal oscillators

- **Round-off noise**: The output of a multiplier has more bits than its inputs. For example, a 16-bit by 16-bit 2's complement multiplication leads to an output word length of 32-bits. To store the output, it has to be re-quantized (the least significant bits are thrown away ... This leads to an error called round-off noise.
- Binary multiplication of two n-bit numbers results in a 2n-bit result. The result of multiplying two Q15 numbers is a 32-bit number in Q30 representation with a redundant sign bit:

```
a = 0.3456783, quantized to Q15: 0.3456726, err: 0.0000057,  bin:0010110000111111
b = 0.2345674, quantized to Q15: 0.2345581, err: 0.0000093,  bin:0001111000000110

c_q30 = a_q15 * b_q15  = 0.0810803          bin: 00000101001100000110101101111010
c_q15                  = 0.0810547

err = diff(c_q30/c_q15)= 0.0000256          bin: 0000101001100000
```

- To map the Q30 result back to the original Q15 format, we skip one of the two sign bit (they are identical anyway) and the lower 15 bits.